BrightLearn

Machine Learning Research Assignment

Four parts. Twelve further questions. One report that shows you understand how machines learn from data — and how to get the data ready first.

Python · pandas · scikit-learn · Jupyter Notebook

Issued 19 October · Due 2 November

2026

Machine Learning Research Assignment

Machine learning is how a computer learns patterns from data instead of following rules written by a person. Before you train your first model you need to understand what machine learning is, the two main ways machines learn, and the work that goes into preparing data so a model can learn from it. This assignment asks you to research those three things, try them for yourself with pandas and scikit-learn, and write up what you found in your own words.

ISSUED	DUE	FORMAT	LENGTH	MARKS
Monday 19 October 2026	Monday 2 November 2026	Written report + notebook	1 500 – 2 500 words	100

Four parts, one report

PART A · 20 MARKS

What is machine learning?

Definitions, real-world examples and the ML workflow

PART B · 25 MARKS

Supervised vs unsupervised

The two main ways machines learn, and how to tell them apart

PART C · 35 MARKS

Preparing the data

Cleaning, encoding, scaling and splitting with pandas and scikit-learn

PART D · 15 MARKS

Further research

Choose five questions and dig one step deeper

+ 5 MARKS — **Presentation and referencing.** Clear structure, your own words, and every source properly cited. See the marking guide at the back of this document.

What you will be able to do when you finish

- Explain machine learning to someone who has never heard of it
- Tell supervised and unsupervised learning apart and name an algorithm for each
- Explain why models need numbers, not text, and encode categorical data
- Scale numeric features and explain which algorithms need it
- Split data into training and test sets without leaking information
- Find your way around the pandas and scikit-learn documentation

Research, don't just read. For every question, try to find the answer in at least two sources, then test it yourself in a notebook wherever you can. The official pandas and scikit-learn documentation counts as a source — and it is usually the most accurate one. Where a question names a function or class, look it up, read what it does, and try it on a few rows of data.

PART A OF 4

What is machine learning?

Definitions, examples and the workflow

20 marks

Research · Written answers · about 400 words

This part's goal. Build the vocabulary. By the end you should be able to explain what machine learning is, where it fits in the bigger picture of AI and data science, and what happens from the moment a problem is defined to the moment a model is used.

#	QUESTION	WHAT TO INCLUDE	MARKS
A1	**In your own words, what is machine learning?**	Explain how it differs from traditional programming. In traditional programming a person writes the rules and the computer applies them to data. In machine learning the computer is given data and the answers, and works out the rules. Include a simple diagram or example that shows the difference.	**4**
A2	**How does machine learning relate to artificial intelligence, deep learning and data science?**	A short explanation of each term and a diagram showing how they overlap or sit inside each other.	**3**
A3	**Define the core terms, each with an example: dataset, feature, label (target), model, training, prediction.**	Use one dataset of your choice (for example learner marks, house prices or customer records) so that all your examples fit together.	**6**
A4	**Find three real-world uses of machine learning in South Africa or elsewhere in Africa.**	For each: what data goes in, what the model predicts, and who benefits. Ideas: fraud detection in mobile money, credit scoring, crop or livestock disease detection, product recommendations, predicting network faults.	**4**
A5	**Describe the typical machine learning workflow from start to finish.**	Problem definition → collecting data → preparing data → choosing and training a model → evaluating it → using (deploying) it. Explain why the data preparation step usually takes the most time.	**3**
		PART A TOTAL	**20**

PART A IN YOUR REPORT

About 400 words plus one or two diagrams.

Write for a classmate who has never heard of machine learning. If they could understand it, you have answered well.

BY THE END OF PART A YOU CAN

- ✓ Explain machine learning without jargon
- ✓ Place ML in relation to AI, deep learning and data science
- ✓ Use the core vocabulary correctly

PART B OF 4

Supervised vs unsupervised

The two main ways machines learn, and how to tell them apart

25 marks

Research · Comparison table · Scenarios

This part's goal. Learn to tell the two apart. It comes down to one question: does the training data include the answers (labels)? The tasks, algorithms and success measures all follow from that.

#	QUESTION	WHAT TO INCLUDE	MARKS
B1	**What is supervised learning?**	What "supervised" means here, what labelled data is, and one everyday example.	**2**
B2	**What is unsupervised learning?**	What the algorithm does when there are no labels to learn from, and one everyday example.	**2**
B3	**Compare the two in a table.**	Rows: goal · type of data needed · typical tasks · example algorithms · how success is measured · one real-world example of each.	**5**
B4	**Explain classification and regression (both supervised).**	How they differ, one example of each, and what kind of value each predicts (a category or a number).	**3**
B5	**Explain clustering and dimensionality reduction (both unsupervised).**	What each is for and one example of each.	**3**
B6	**For each scenario, say whether it is supervised or unsupervised, then classification, regression or clustering, and why.**	(a) Predicting a house's selling price from its size and suburb. (b) Grouping customers by spending habits, with no groups defined upfront. (c) Deciding whether an SMS is spam. (d) Predicting a learner's final mark from term marks. (e) Finding groups of products with similar sales patterns. One or two sentences each.	**5**
B7	**Look up these scikit-learn classes. For each, say whether it is supervised or unsupervised and what it is used for.**	`LinearRegression`, `LogisticRegression`, `DecisionTreeClassifier`, `KNeighborsClassifier`, `KMeans`, `PCA`. Use the User Guide at scikit-learn.org and give the module each one lives in (e.g. `sklearn.cluster.KMeans`).	**5**
		PART B TOTAL	**25**

PART B IN YOUR REPORT

About 500 words, including your comparison table.

Make the B3 table one a classmate could revise from. **Optional extra (no marks):** a paragraph each on semi-supervised and reinforcement learning, with an example.

BY THE END OF PART B YOU CAN

- ✓ Tell supervised and unsupervised learning apart instantly
- ✓ Match a business problem to classification, regression or clustering
- ✓ Name a scikit-learn algorithm for each

PART C OF 4

Preparing the data

Cleaning, encoding, scaling and splitting with pandas and scikit-learn

35 marks

Research 25 marks · Notebook 10 marks

This part's goal. Real data is never ready. It has missing values, text where numbers should be, and columns on wildly different scales. Research each step below, find the pandas or scikit-learn tool that does it, and then do it yourself in a notebook. This is the part of the assignment you will use most in the rest of the course.

STEP	RESEARCH QUESTIONS · AND THE PANDAS / SCIKIT-LEARN TOOLS TO LOOK UP	MARKS
C1 **Explore and clean**	What should you check first when you open a new dataset? How do you find missing values, duplicate rows and columns with the wrong data type (for example numbers stored as text)? LOOK UP `df.info()` `df.describe()` `df.isnull().sum()` `df.duplicated()` `df.drop_duplicates()` `df.astype()`	2
C2 **Handle missing values**	What are the options: drop the rows or columns, or fill them with the mean, median, mode or a constant? When is each appropriate? Why might dropping rows be a bad idea on a small dataset, and why use the median rather than the mean when a column has outliers? LOOK UP `df.dropna()` `df.fillna()` `sklearn.impute.SimpleImputer`	3
C3 **Encode categorical data**	Why can a model not work with text like "Gauteng" or "Prepaid"? Explain label encoding, ordinal encoding and one-hot encoding, and when to use each. What is the difference between nominal and ordinal categories, and why does it matter which encoder you pick? What problem does one-hot encoding cause on a column with hundreds of unique values? LOOK UP `pd.get_dummies()` `sklearn.preprocessing.LabelEncoder` `OrdinalEncoder` `OneHotEncoder`	6
C4 **Scale numeric features**	Why does a column like salary (in thousands) next to age (in tens) cause problems for some models? Explain standardisation (z-score) and normalisation (min–max) and when you would use each. Which algorithms need scaling (for example KNN, logistic regression, SVM) and which do not (decision trees, random forests)? What is `RobustScaler` for? LOOK UP `sklearn.preprocessing.StandardScaler` `MinMaxScaler` `RobustScaler`	6
C5 **Split into training and test sets**	Why do we hold back part of the data, and what is a typical split? What do `random_state` and `stratify` do? What is data leakage, and why must you fit the scaler on the training data only and then transform the test data? Explain the difference between `fit()`, `transform()` and `fit_transform()`. LOOK UP `sklearn.model_selection.train_test_split`	4

STEP	RESEARCH QUESTIONS · AND THE PANDAS / SCIKIT-LEARN TOOLS TO LOOK UP	MARKS
C6 **Deal with outliers**	How do you spot outliers (boxplots, the IQR rule, z-scores)? Should you always remove them? Give one case where an outlier is a data entry mistake and one where it is real and important. LOOK UP `df.quantile()` `df.clip()` `seaborn.boxplot()`	2
C7 **Engineer features**	What is feature engineering? Give two examples, such as taking the month out of a date or calculating age from a year of birth. How can a correlation matrix help you decide which features to keep? LOOK UP `pd.to_datetime()` `.dt.month` `df.corr()`	2
	PART C RESEARCH TOTAL	25

PART C PRACTICAL · 10 MARKS

Do it in a notebook.

Use any small dataset — the class practice dataset, the Titanic dataset from `seaborn.load_dataset('titanic')`, or a CSV from Kaggle. Create a Jupyter notebook that:

1. Loads the data with pandas and shows its shape, data types and missing values
2. Fills or drops missing values, and says why
3. Encodes at least one categorical column
4. Scales at least two numeric columns
5. Splits the data 80 / 20 into training and test sets — scaling after the split, not before

Add a markdown cell above each step explaining what you did and why.

BY THE END OF PART C YOU CAN

- ✓ Turn a raw CSV into a model-ready table
- ✓ Choose the right encoder for the right kind of column
- ✓ Choose the right scaler and know when scaling is unnecessary
- ✓ Split data without leaking the test set into training

Five common mistakes — check your notebook for them before you submit.

1. Fitting the scaler on the whole dataset before splitting, so the test set leaks into training.
2. Using `LabelEncoder` on a nominal column like province, which tells the model that "Limpopo" is bigger than "Gauteng".
3. Dropping every row with a missing value and losing half the data in the process.
4. Scaling or encoding the target column along with the features.
5. Cleaning the training data but forgetting to apply exactly the same steps to the test data.

PART D OF 4

Further research

Choose five questions and dig one step deeper

15 marks

Any 5 questions · 3 marks each · 150 – 250 words each

This part's goal. The questions below go one step beyond the basics. Choose any five. For each, write 150 – 250 words in your own words and cite at least one source. Pick the ones that interest you — these are the topics you will meet first when you start training models in class. Finished early? AutoML, MLOps (and MLflow) and large language models make good optional extras.

D1

Overfitting and underfitting

What are they, what causes each, and how does a train/test split help you spot them? Sketch what each looks like on a graph.

D2

Measuring a model

Research accuracy, precision, recall, F1-score and the confusion matrix (classification), and MAE, RMSE and R^2 (regression). If only 1 in 100 transactions is fraud, why is accuracy a poor metric, and which would you use instead?

D3

Cross-validation

What is k-fold cross-validation and why is it more reliable than a single train/test split? Look up `cross_val_score` in `sklearn.model_selection`.

D4

Parameters vs hyperparameters

What is the difference? Give an example of each for a decision tree or KNN. What does `GridSearchCV` do, and why can it be slow?

D5

Decision trees and random forests

How does a decision tree make a decision? Why does a random forest usually do better than a single tree? Why do neither of them need feature scaling?

D6

K-means clustering

How does the algorithm work, step by step? How do you choose the number of clusters (the elbow method), and why does K-means need scaled data?

D7

Class imbalance

What is it, why does it mislead models, and what can you do about it (resampling, class weights, SMOTE)? Where does it show up in fraud, churn or medical data?

D8

Machine learning vs deep learning

What is a neural network, in plain language? When would you reach for one instead of scikit-learn, and what does it cost you in data and computing power?

D9

Pipelines

What are `sklearn.pipeline.Pipeline` and `sklearn.compose.ColumnTransformer`? How do they stop you from leaking data or forgetting a preprocessing step when you move from training to test data?

D10

Bias and fairness

How can a model be unfair even when nobody intended it to be? Find one documented example (credit scoring, hiring, facial recognition) and explain how the training data caused it.

D11

Data protection and POPIA

What is POPIA, and what does it mean for collecting and using personal data to train models in South Africa? What is the difference between personal data and anonymised data?

D12

Careers in machine learning

Research one role: data analyst, data scientist or machine learning engineer. What skills, tools and qualifications do South African job adverts ask for, and what does the role do day to day?

MARKING GUIDE

How marks are allocated

What earns the marks, and what loses them

100 marks

Report 90 · Notebook 10

How we mark. Each part is marked against the questions on its page, up to the marks shown next to each question. Use the descriptors below to check your own work before you submit.

SECTION	WHAT WE LOOK FOR	MARKS
Part A What is machine learning?	Clear definitions in your own words, correct use of the vocabulary, relevant local examples	**20**
Part B Supervised vs unsupervised	Accurate comparison table, scenarios classified correctly with reasons, correct scikit-learn classes and modules	**25**
Part C — research Preparing the data	Each step explained with the right pandas or scikit-learn tool, and sound reasoning about when to use which option	**25**
Part C — notebook Preparing the data	Runs cleanly, every step present and explained, scaler fitted on the training set only	**10**
Part D Further research	Five questions answered with depth beyond a definition, each with at least one source	**15**
Presentation and referencing	Logical structure, readable writing, consistent referencing, no copied text	**5**
Total		**100**

What a strong answer looks like

	EXCELLENT · FULL MARKS	GOOD · MOST MARKS	NEEDS WORK · FEW MARKS
Understanding	Explains the idea in your own words, with an example that shows you could apply it to new data	Correct explanation, but leans on textbook wording or a borrowed example	A definition copied or paraphrased from one source, with no example
Research depth	Draws on two or more sources, notes where they differ, and checks claims against the official documentation	One reliable source per question, used correctly	A single web page or chatbot answer, unverified
Use of pandas and scikit-learn	Names the right function or class, the module it lives in, and shows it running on real rows in the notebook	Names the right tool and describes what it does	Tools missing, misnamed, or used on the wrong kind of column
Reasoning	Says *when* and *why* to use one option over another — median over mean, one-hot over label encoding, scaling after the split	Describes the options correctly but does not compare them	Lists steps without explaining the choices behind them
Presentation and referencing	Clear headings by question number, consistent Harvard or APA references, a tidy notebook that runs top to bottom	Readable and referenced, with small inconsistencies	Hard to follow, references missing or incomplete, notebook does not run

FINAL STEP

Submitting your work

Your research, your words

Due 2 November

Monday · 11:59pm

How to submit. Submit two files: your report as a PDF (or Word document) and your notebook as an `.ipynb` file. Name them `Surname_Name_ML_Research.pdf` and `Surname_Name_ML_Research.ipynb`, and send them the way your facilitator has set up for the class.

YOUR REPORT SHOULD CONTAIN

- A cover page with your name, the date and the course
- A contents page
- Parts A to D, using the question numbers (A1, B3, C4 …) as headings
- A reference list in Harvard or APA style, with at least five sources
- A short declaration of any AI tools you used, and for what

YOUR NOTEBOOK SHOULD

- ✓ Run from top to bottom without errors
- ✓ Have a markdown cell explaining each step
- ✓ Use pandas and scikit-learn (plus matplotlib or seaborn for plots)
- ✓ Keep its outputs visible — do not clear them before submitting

Using AI tools (ChatGPT, Claude, Gemini, Copilot). You may use them to help you understand a topic or explain an error message, but every answer must be written in your own words and checked against the official documentation. List the tools you used in your declaration. Text copied straight from a chatbot or a website will not earn marks.

Recommended resources

- **scikit-learn User Guide and "Getting Started"** — scikit-learn.org. Your first stop for every class named in this assignment.
- **pandas: "10 minutes to pandas" and the User Guide** — pandas.pydata.org
- **Google Machine Learning Crash Course** — developers.google.com/machine-learning (free, with short videos)
- **Kaggle Learn** — kaggle.com/learn: Intro to Machine Learning, Pandas, Intermediate Machine Learning (free, hands-on)
- **StatQuest with Josh Starmer** (YouTube) — clear, visual explanations of almost every concept in Parts B and D
- **Aurélien Géron, *Hands-On Machine Learning with Scikit-Learn, Keras and TensorFlow*** — chapters 1 and 2 cover Parts A and C

Questions?

Ask in class or email your facilitator before Monday 2 November 2026.

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